SOP-PRODUCTION PROCESS

[CONFIDENTIAL]
[UTTHUNGA]
[IndiQube, South Summit, South End Rd, Vijayarangam Layout, Bengaluru, Karnatake 560004]

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CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Prepared By Date	Reviewed By Date	Approved By Date
1	1.0	New Process	Initial	Siva prasad 21-Oct-24		
2	Issue B, V1.0	Reviewed and approved			Anitha. N 24-Oct-24	Santosh. V 25-Oct-24
3	Issue B, V1.1	All sub-section modified	Details added	Santosh V 17-Apr-24	Ranjit, Jegajith, Dharma 17-Apr-24	Sanmangala 17-Apr-24

1 Purpose

This SOP defines the production management process when manufacturing is outsourced to a vendor. It ensures that:

The company retains full production ownership.

The EMS (Electronics Manufacturing Services) vendor adheres to approved design specification.

The production process meets established quality standards and deadlines.

The product quality aligns with the Quality Control (QC) documentation.

2 Scope

This SOP applies to outsourced production activities for products designed and developed by the company as well as turnkey projects solely undertaken for production. It involves all internal teams, including Testing, Procurement, Production, Quality Assurance, and Vendor Management, as well as the third-party EMS vendor responsible for production. Additionally, it includes the company's responsibility to define and implement quality processes in accordance with production standards.

For In-house production, refer SOP manufacturing and assembly

3 Abbreviations

- PDS: Product Design Specifications
- BOM: Bill of Materials
- CAD: Computer Aided Design
- CNC-Computer numerical control
- D&D: Design and Development
- ECR: Engineering Change Request
- ECO: Engineering Change Order
- CAPA: Corrective and Preventive Action
- Hex: Hexadecimal
- QC: Quality Control
- EMS: Electronics Manufacturing Services
- AQL-acceptable quality limit
- ASL-Approved Supplier List
- MOQs-Minimum order quantities
- MRP-Material Requirement Planning
- RCA-Root Cause Analysis
- ISO: International Organization for Standardization
- NDAs: Non-Disclosure Agreements

4 Terms and Definitions

4.1 Vendor Audit: A systematic examination of a vendor's operations, financial stability, past performance, certifications, production capacity, and regulatory compliance.

4.2 Prototype: An initial model or sample of a product used to evaluate and validate design specifications.

4.3 Risk Register: A tool used to document and monitor potential risks and their mitigation strategies.

4.4 Corrective and Preventive Action (CAPA): A process to identify, address, and prevent the recurrence of quality issues.

4.5 Engineering Change Request (ECR): A formal request to modify a product's design.

4.6 Engineering Change Order (ECO): A document that authorizes and details the implementation of a design change.

4.7 Quality Control (QC): Processes and procedures to ensure products meet specified quality standards.

4.8 Bill of Materials (BOM): A comprehensive list of materials, components, and assemblies required to build a product.

5 Responsibilities

Procurement Team: The Procurement Team is responsible for identifying potential vendors, facilitating audits, and negotiating contracts while ensuring compliance with confidentiality agreements. They manage timely delivery of materials, verify receipt and quality, and maintain procurement documentation to ensure consistency and accuracy.

Vendor Assessment Team: This team conducts audits to assess vendors' financial stability, certifications, and production capacity, reviews submitted samples and makes approval or rejection decisions based on audit results to ensure that vendors meet quality and regulatory standards.

Design Team: The Design Team provides detailed Product Design Specifications (PDS) and Bill of Materials (BOM), addresses vendor queries about production feasibility, participates in design reviews and validation, documents change through Engineering Change Requests (ECRs), and updates relevant documentation to reflect any revisions.

Production Manager/Production Supervisor: The Production Manager/Production Supervisor coordinates design handoff meetings, inspects prototypes, ensures vendors have comprehensive production plans, establishes communication channels for updates, oversees quality control processes, and verifies compliance in packaging and shipment procedures.

Project Manager: The Project Manager manages design changes and coordinates with vendors to adjust timelines, oversees quality control through CAPA processes, ensures continuous improvement, and establishes acceptance criteria for final product testing before deployment while collecting feedback for future enhancements.

Legal: The legal advisor comes up with terms and conditions, recommends on regulatory and compliance requirements

6 RACI

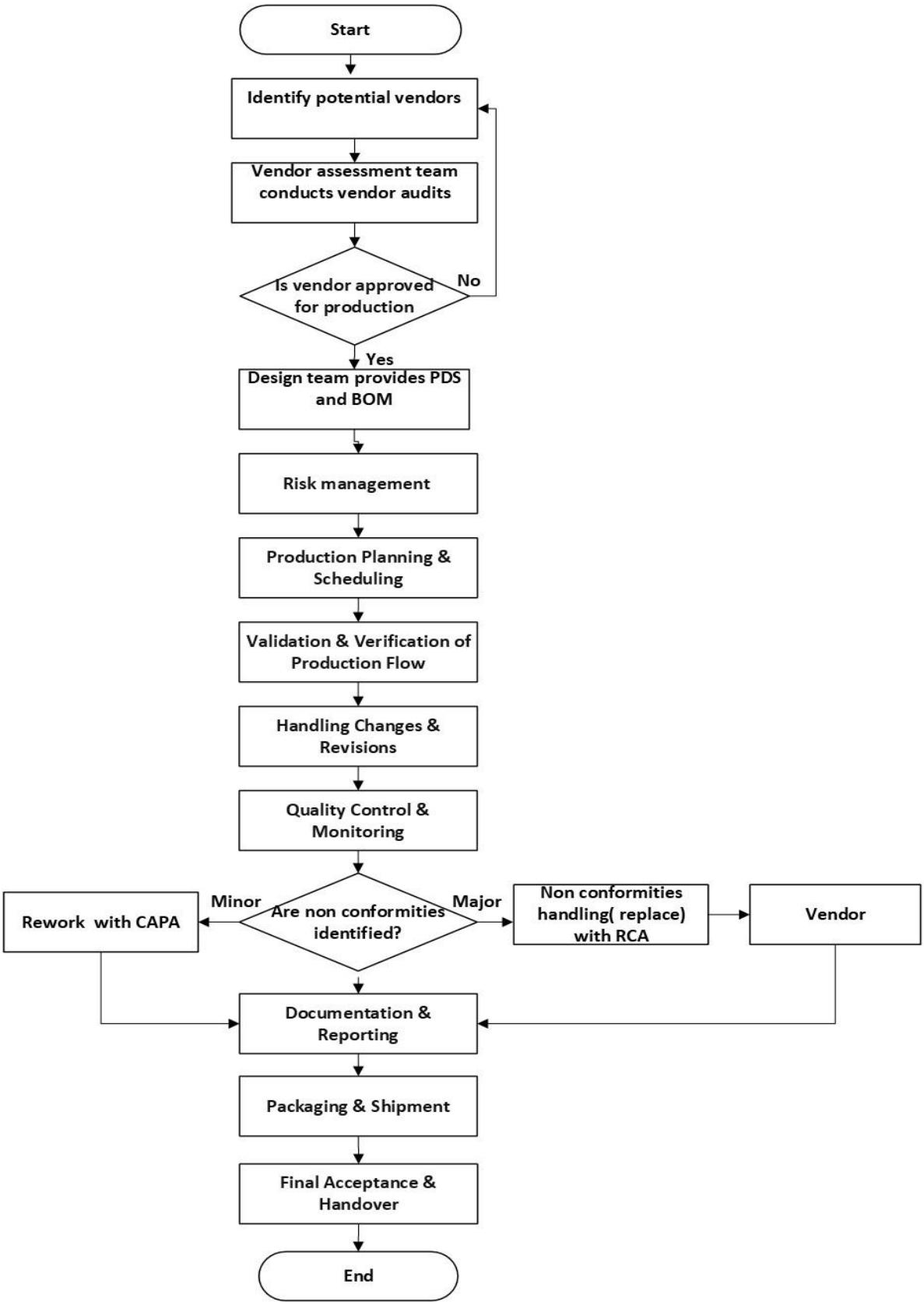
Process	Responsible	Accountable	Consulted	Informed
Vendor Selection & Approval	Procurement Team, Vendor Assessment Team	Procurement Team	Production Team, Design Team, Legal	Stakeholders, Project Team
Release for Production	Production Manager/Production Supervisor	Production Manager/Production Supervisor, Design Team,	Hardware lead, Vendor	Procurement Team, Project Manager
Risk Management	Project Team, Production Team	Project Manager	Vendors, Procurement Team	Design Team, Senior Management
Production Planning	Production Manager/Production Supervisor	Project Manager	Procurement Team, Vendor	Design Team, Project Team
Manufacturing	Vendor	Production Manager/Production Supervisor	Design Team, Quality Control team	Project Team, Quality Assurance Team
Handling Changes and Revisions	Production Manager/Production Supervisor	Project Manager	Vendors, Production Team	All stakeholders involved in the change
Validation & Verification of Production Flow	Vendor	Production Manager/Production Supervisor	Procurement Team, EMS Vendor, Vendors	Project Team, Quality Assurance Team
Quality Control of finished goods	Production Manager/Production Supervisor	Project Manager	Vendor, Quality Assurance Team	Design Team, Procurement Team
Packaging, Logistics & Shipping	Production Team, Vendor	Production Manager/Production Supervisor	Logistics Provider, Vendor	Procurement Team, Client
Final Acceptance & Handover	Production Manager/Production Supervisor	Project Manager	Client, Vendor	Production Team, Design Team, Senior Management

7 SIPOC

Supplier	Input	Process and Control		Output	Customer
		Process	Control		
Vendor Assessment Team	Potential vendors, vendor profiles, audit checklists	Vendor Selection & Approval	Vendor audit, assessment of samples, Quality control checks, QMS certificates	Approved/ rejected vendor list, signed contracts	Production Team, Project Team
Production Manager/Production Supervisor	Technical documentation, Manufacturing instructions, QC Guidelines, Supply Chain & Logistics	Release for production	Design handoff meeting,	Acceptance by Vendor	Vendor
Project Manager	Identified risks, historical data, risk register	Risk Management	Risk assessment meetings, mitigation strategies, monitoring risk register	Updated risk register, mitigation plans	Project Team, Production Team, Procurement Team
Production Manager/Production Supervisor	Production Schedule & Supply Chain Plan	Production Planning	Production plan and milestones, timely material delivery	Production schedule, Material requirement plan, resource allocation plan	Vendors, Procurement Team

Vendor	Input Material, Production plan	Manufacturing	Pilot run validation, Quality Control Checks	Manufactured goods	Production Manager/Production Supervisor
Project Manager	Engineering Change Requests (ECRs) Engineering Change Orders (ECOs)	Handling Changes & Revisions	Approved changes to process, cost, Vendor Notification,	Updated design documents, revised production schedule	Production Manager/Production Supervisor , Vendors,
Hardware Design Team	Initial designs, test plans, production prototypes	Validation & Verification of Production Flow	Testing, simulation, prototype validation, functional and stress tests	Validated product prototypes, test reports	Quality Assurance Team, Project Team
Production Team	Production outputs, quality inspection criteria	Quality Control & Monitoring	In-process inspections, CAPA processes, regular quality data reviews	Quality reports, CAPA records, accepted/ rejected products	Project Team, Procurement Team, Customers
Production Team & Vendor	Finished products, packaging standards	Packaging, Logistics & Shipment	Packaging according to standards and tracking	Packaged and labeled products, shipment tracking info	Customers, Logistics Providers, Project manager
Project Manager	Completed product, final inspection reports	Final Acceptance	Client-side testing, review feedback & document transfer	Final product acceptance sign-off, project handover documentation, Customer feedback	Clients, Senior Management

8 Overall Process Flow Chart



9 Process Description

9.1 Supply Chain management

9.1.1 Procurement team identifies vendors as per Procurement SOP for Raw Materials and Components

a. Review the product design and determine the raw materials and components required.

b. Create a list of specifications, including part numbers, quantities, lead times, and quality standards.

c. Prioritize critical materials/components based on product functionality and production schedules.

d. Research and Shortlist Potential Vendors

- Conduct market research to identify potential suppliers
- Evaluate vendor reputation, financial stability, delivery performance, and certifications based on supplier evaluation process

e. Evaluate Manufacturing Partners' Production Capabilities

- Identify critical processes for production
- Consider the need for specialized production techniques or certifications

f. Evaluate Manufacturing Partner's Capabilities:

- Evaluate the vendor as per Procurement SOP
- Research potential manufacturing partners that have the necessary equipment, technology, and experience
- Assess their production capacity to ensure they can handle the volume and meet delivery deadlines.
- Evaluate their flexibility in handling design changes, prototyping, and scaling up production.
- Use Onsite supplier assessment checklist UTSFPURTMP18.xls

g. Assess Manufacturing Processes and Quality Control

- Supplier Onsite inspection and In-process and product inspection as per Procurement SOP.
- Conduct Pilot Runs or Prototyping:
- Run small-scale production or pilot runs to test their ability to meet quality and performance standards.
- Assess the prototype quality, including fit, finish, and functionality.

- Review the timeline, cost structure, and lead times from the pilot runs to ensure they align with expectations
- Engineering Assessment: Verify Suppliers' Ability to Meet Design Specifications and Quality Requirements
- Finalize the engineering requirements for each component, including material properties, tolerances, performance characteristics, and any required certifications.
- Prepare detailed drawings, CAD models, or specifications for the suppliers, including testing and inspection criteria.

h. Evaluate Supplier's Technical Capabilities

- Assess whether the supplier has the necessary technical expertise to meet the product's design requirements (e.g., high-precision machining, custom PCB design).
- Verify their design and engineering team's qualifications and experience in handling similar projects.
- Confirm that they have access to the right technology and equipment (e.g., CNC machines, 3D printers, laser cutters) to meet the specifications.

i. Review Testing and Quality Assurance Protocols

- Review their in-house testing and validation procedures (e.g., tensile testing, environmental testing, functionality testing).
- Ensure they can meet any required industry standards or certifications (e.g., CE marking, UL certification, RoHS compliance).
- Assess their ability to perform design validation and verification (e.g., prototypes, pre-production runs).

j. Audit Supplier's Quality Management Systems (create checklist)

- Review the supplier's quality, security and environment management systems (e.g., ISO 9001 certification, internal auditing processes).
- Ensure that they have robust traceability and documentation practices for all components.
- Evaluate their ability to perform root cause analysis and corrective actions in case of quality issues.

k. Evaluate Prototypes and Production Samples

- Request and evaluate engineering prototypes or samples before moving to full-scale production.
- Perform comprehensive testing of prototypes to verify that the supplier can meet the design specifications and quality requirements.

- Assess the supplier's responsiveness to feedback and their ability to adjust based on test results.
- Use Vendor evaluation form UTSFPURTMP05

9.2 Release for production

9.2.1 Design team provides the Production Manager/Production Supervisor the following technical documentation for production and Production Manager/Production Supervisor communicates it to the Vendor

Category	Required Documents & Information
Technical Documentation	BOM, CAD Drawings, Firmware & software files, Schematics & Circuit Diagrams
Manufacturing Instructions	Assembly Instructions, Testing Procedures, Packaging Details
Quality & Compliance	QC Standards, Compliance & Certifications, Tolerance & Acceptance Criteria
Supply Chain & Logistics	Supply Chain & Logistics Information, Procurement & Lead Time Information, Shipping & Logistics Coordination details

a. Bill of Materials (BOM)

- List of all components, subassemblies, and raw materials. Use Bill of Material template UTSFPURTMP10 & UTSFPURTMP13
- Unit cost estimates
- Component tolerances & material specifications

b. CAD Drawings

- Mechanical Drawings
- PCB Layout Files (Gerber files for circuit boards)
- Exploded Views (Assembly structure & component arrangement)
- Dimensional & Tolerance Specifications

c. Firmware & Software Files

- Embedded system code & firmware (C, Python, or compiled HEX/BIN files)
- Microcontroller bootloader & communication protocols
- API documentation for connectivity
- Use for release for production UTSFPURTMP28
- Manufacturing Instructions
- Ensure the security measures are in place to ensure that files are secure during transmission and post receive by vendor. Implement appropriate measures like Digital signatures, secure files transfer, access control, validation of receipt, secure hardware elements, Cryptographic Hash

Validation, Secure Boot / Trusted Platform Module, legal controls, Encrypted Formats as appropriate.

d. Schematics & Circuit Diagrams

- Electronic design schematics
- Power distribution, signal paths, and component placement

e. Assembly Instructions

- Step-by-step process for putting together components
- Sequence of assembly (mechanical first, then electronic, etc.)
- Soldering, welding, or bonding instructions

f. Testing Procedures

- Functional testing scripts (automated & manual)
- Pass/fail criteria for each stage
- Environmental & stress testing requirements

g. Packaging Guidelines

- Final product packaging (box design, inserts, padding materials)
- Retail labeling, barcodes, and QR codes
- Shipment packaging for bulk orders

9.2.2 Quality & Compliance Requirements

a. Quality Control (QC) Standards

- Incoming material inspection procedures Refer SOP – Incoming Material Inspection UT/PS/IGI/SOP
- In-process & final product quality checks
- Sampling plan & rejection criteria (AQL standards)

b. Compliance & Certifications

- Industry standards (CE, FCC, ATEX, RoHS, ISO 9001, UL, etc.)
- Product safety requirements (shock, fire, electrical hazard compliance)
- Country-specific regulations (FDA for medical devices, BIS for India, etc.)

c. Tolerance & Acceptance Criteria

- Mechanical tolerances
- Electronic tolerances (Voltage, current limits)
- Cosmetic defect tolerances (scratches, color variations)

d. Supply Chain & Logistics Information

- Approved Supplier List (ASL)
- List of preferred suppliers for critical components
- Alternatives in case of shortages

e. Procurement & Lead Time Information

- Component availability & expected delivery times

- Minimum order quantities (MOQs) for bulk purchases

f. Shipping & Logistics Coordination details

- Incoterms (FOB, CIF, DDP)
- Storage conditions (humidity, temperature limits for electronics)
- Labeling & tracking requirements

9.3 Risk Management

a. The Production Manager/Production Supervisor with project manager with inputs from vendor and procurement does risk assessment as per SOP - Risk & Opportunity Management UT/IM/RMP/SOP and comes up with mitigation and contingency plan as applicable.

b. Acts as per the mitigation and contingency plan.

9.4 Production Planning

9.4.1 Production Manager/Production Supervisor creates the Production Schedule & Supply Chain Plan and shares with the Vendor. As part of planning does the following activities.

a. Set Production Milestones: Create a timeline with clear milestones for prototyping, pilot runs, full-scale production, and product testing.

b. Define Lead Times: Consider lead times for raw material procurement, component delivery, and manufacturing time.

9.4.2 Scheduling

a. Vendor come up with daily, weekly, and monthly production schedules to optimize equipment utilization and minimize downtime.

b. Schedule buffer time for maintenance activities and unplanned disruptions.

c. Ensure production schedules are updated based on demand fluctuations or resource availability.

9.4.3 Material Requirement Planning (MRP):

a. Generate material requirements based on the Bill of Materials (BOM) and production plan.

b. Ensure timely procurement of materials as per the Procurement SOP.

c. Implement Just in Time (JIT) principles to minimize stock levels, reduce storage costs, and ensure timely availability of parts.

d. Identify potential risks in the supply chain (e.g., shipping delays, material shortages) and plan for contingencies

e. Monitor material consumption and adjust procurement accordingly.

f. Ensure buffer stock is maintained as per the plan

9.4.4 Lean Manufacturing Implementation

a. **Optimize Production Flow:** Apply Lean principles such as waste reduction, continuous improvement, and value stream mapping to streamline production processes.

b. **Cycle Time Reduction:** Focus on reducing production cycle times by eliminating non-value-added activities.

c. **Workforce Training:** Train operators and supervisors on Lean practices to continuously improve efficiency

9.5 Manufacturing Process Execution

9.5.1 Pilot Runs: The vendor produces limited quantities to validate the manufacturing process, verify product design, and identify issues early in the production cycle.

9.5.2 Full-Scale Production: Vendor transitions from pilot runs to mass production once the process is validated and optimized based on the approval from Production Manager/Production Supervisor

9.5.3 Process Monitoring: Use automated monitoring systems to track production speed, material usage, and quality control.

9.6 Handling Changes and Revisions

9.6.1 Project Manager communicates all design changes through Engineering Change Requests (ECRs) ENGINEERING CHANGE REQUEST FORM - ECR or Engineering Change Orders (ECOs) follow SOP – Engineering Change Request Management UT/SD/ECRM/SOP

9.6.2 Informs the Production Manager/Production Supervisor of any design changes and adjusts production timelines, as necessary.

9.6.3 Production Manager/Production Supervisor assess the scope of design changes to determine if the production needs to submit a new prototype or perform validation runs before resuming production.

9.6.4 Ensure all design changes are clearly documented and approved before implementation.

9.6.5 Vendors communicate any potential impact on handling charges due to design changes to Production Manager/Production Supervisor

9.6.6 Monitor and track all revisions to ensure they are accurately reflected in production processes.

9.6.7 Production Manager/Production Supervisor coordinates with the vendor to update any affected documentation, such as bill of materials (BOM) and assembly instructions.

9.6.8 Review and approve any additional costs associated with design changes and revisions.

9.6.9 Establish a process for resolving any discrepancies or issues that arise from design changes.

9.7 Verification and Validation of Production Flow

9.7.1 Vendor defines scope and requirements as per inputs provided.

a. Document the process flow
b. Check for quality gates
c. Test each step of production flow and perform dry runs
d. Qualify equipment

9.7.2 Vendor Validation:

a. Confirm end-to-end flow achieves intended results
b. Run production in controlled scenarios to validate output
c. Collect and analyze process data to ensure repeatability and reliability

9.7.3 Create V&V report

9.8 Quality Control (QC) of Finished Goods

9.8.1 In-Process Quality Control:

a. Vendor Quality Control team conducts Visual and Functional Inspections: Perform in-process checks during production to ensure that parts and assemblies meet design specifications.
b. Sampling Inspection: Randomly sample finished goods and test them against performance and compliance criteria.
c. Non-Destructive Testing: Use testing methods to ensure product integrity without damaging it.

9.8.2 End-of-Line Testing:

a. Vendor Quality Control team performs comprehensive tests on the finished product to verify functionality (e.g., electronic testing, pressure tests for seals).
b. Ensure compliance with relevant standards
c. In case vendor must ship directly, the final inspection shall be done at vendor location as per agreement.
d. Incoming Inspection
   - Production QC team performs inspection on incoming goods as per SOP – Incoming Material Inspection
   - Generate report using Incoming Goods Inspection Report UTSDHPTMP05

9.8.3 Non-Conformance and Corrective Action

- The Quality Control team documents non-conformances SOP - Control of Non-Conforming Product UT/PS/CNP/SOP using the Non-Conformance Report Template UT/PS/CNP/TMP/02.
- Conduct root cause analysis (RCA) and document the RCA report.
- Implement corrective and preventive actions and monitor effectiveness

9.9 Packaging, Logistics, and Shipping

9.9.1 Ensure shipping is done by following the Packaging using Packing & Shipment SOP UT/PS/PAS/SOP. Production team takes care of the shipping

after receiving from vendor or the Vendor does the shipping as per agreement.

9.9.2 Labeling: Include necessary labels, including part numbers, barcodes, safety instructions, and any regulatory compliance marks as per SOP - Packing & Shipment UT/PS/PAS/SOP

9.9.3 Include necessary certificates and test reports with each shipment.

9.9.4 Shipping and Delivery:

a. Procurement team selects logistics partners based on approved vendor lists.
b. Define and monitor delivery timelines to ensure on-time delivery.
c. Ensure adequate measures like insurance, appropriate packaging and qualified logistics are taken to ensure delivery failures or damages are taken care of.

9.10 Acceptance

9.10.1 Acceptance Criteria:

a. Inspection and Sign-Off: Upon receipt, customers should verify that the product matches the agreed-upon specifications and meets performance standards.
b. Product Performance Guarantee: Establish clear performance criteria (e.g., product lifecycle, warranty periods).
c. Return and Refund Policies: Ensure clear policies are in place in case of defective products or customer dissatisfaction.

9.10.2 Post-Shipment Feedback Loop:

a. Collect Customer Feedback: After shipment, request feedback on product performance and customer satisfaction.
b. Analyze and Improve: Use the feedback to identify areas for product and process improvement.

10 Information Security requirements

10.1 Suppliers shall be assessed based on adherence to information security requirements specified by the Utthunga.

10.2 Supplier shall sign the NDA with the organization if required. NDA is applicable for all service-based suppliers, it is also applicable when confidential data such as specifications, drawings, designs, patterns is to be shared to the suppliers (subcontractors, consultants & testing)

10.3 ICT (Information Communications Technology) based suppliers shall also be monitored on agreed SLA's (Service Level Agreements)

11 References

11.1 ISO 9001 Clauses 8.1

11.2 ISO 27001 Clauses 8.1, Controls A.14

11.3 ISMS Control Policies

11.4 SOP – Procurement

12 Metrics

12.1 On-Time Delivery Rate

12.2 First Pass Yield

12.3 Vendor Approval Rate

12.4 Risk Mitigation Success Rate

13 Annexures

Nil

14 Forms

14.1 New Vendor Registration Form UT/SF/PRO/TMP/03

14.2 Approved Vendor List UT/SF/PRO/TMP/04

14.3 Vendor Evaluation Form UT/SF/PRO/TMP/05

14.4 Goods Receipt and Inspection Report UT/SF/PRO/TMP/06

14.5 Bill of Materials UT/SF/PRO/TMP/10

14.6 Materials Dispatch Checklist UT/SF/PRO/TMP/11

14.7 Incoming Inspection Report UT/SF/PRO/TMP/12

14.8 Supplier Evaluation UT/SF/PRO/TMP/14

14.9 Purchase Order Checklist UT/SF/PUR/TMP/20

14.10 Quotation Request Form UT/SF/PUR/TMP/21

14.11 Engineering Change Request Form UT/PS/CMP/TMP/06